

BEHAV3D Tumor Profiler to map heterogeneous cancer cell behavior in the tumor microenvironment

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eLife Assessment

This study represents a potentially **useful** tool for extracting quantitative data from intravital microscopy directed at in vivo cancer models. In general, this is an area of interest as accessible non-proprietary tools are needed and some evidence of the tool's utility is provided. However, the work in its current form is **incomplete** as it is heavily reliant on proprietary software to segment, track, and correct the data. In addition, there are significant reservations regarding the methods used to produce statistics in the software, limiting its applicability and the potential advance over other approaches.

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Abstract

Intravital microscopy (IVM) enables live imaging of animals at single-cell level, offering essential insights into cancer progression. This technique allows for the observation of single-cell behaviors within their natural 3D tissue environments, shedding light on how genetic and microenvironmental changes influence the complex dynamics of tumors. The complexity of data generated by IVM often surpasses the capabilities of conventional analyses accessible to biomedical scientists, thereby neglecting single-cell heterogeneity and limiting the exploration of microenvironmental influences on cellular behavior without bias. To address this challenge, here we introduce BEHAV3D Tumor Profiler (BEHAV3D-TP), a user-friendly computational framework designed for the comprehensive analysis of single tumor cell behaviors and their interactions with the tumor microenvironment (TME). BEHAV3D-TP facilitates unbiased profiling of cancer cell dynamics without requiring advanced computational expertise. Here, we apply BEHAV3D-TP to study diffuse midline glioma (DMG), a highly aggressive pediatric brain tumor characterized by invasive growth. Our analysis reveals that distinct migratory behaviors of DMG cells correlate with specific TME components such as tumor-associated macrophages and vasculature. This approach, initially aimed at uncovering tumor invasive patterns and their interactions with the TME, holds

promise for understanding additional cancer cell behaviors like intravasation and metastasis. BEHAV3D-TP represents a significant advancement in democratizing the analysis of heterogeneous cancer cell behaviors and their TME interactions, providing accessible computational insights into tumor dynamics.

Introduction

Cancer is a complex disease driven by a sequence of genetic and microenvironmental changes. As a result, tumors comprise many different cell populations that are in constant evolution¹. This tumor heterogeneity has important biological and clinical implications as it differentially promotes key characteristics of individual cancer cells, including proliferation, invasion, and drug resistance². Yet, we are far from fully understanding all the dynamic and bi-directional interactions in heterogeneous tumors, and how these interactions influence cancer cell behavior at the single-cell level. Emerging single-cell technologies and analysis tools provide a new opportunity to profile individual cells within tumors. For example, recent spatial transcriptomic and proteomic profiling provides evidence for the existence of multiple distinct microenvironmental niches within one tumor³. However, these techniques can only provide static information and make use of thin tissue sections that are only one cell layer thick. To investigate the real-time behavior of individual cells intravital microscopy (IVM) has been developed to study tumor cell behavior evolution over time but also in their three-dimensional native tissue environment.

IVM is a technique used to capture both spatial and temporal information at a single-cell level in living animals. Over the past years, it has provided many new insights into the dynamics of tumor heterogeneity and the development of therapy resistance^{4–14}. For example, intravital imaging of mouse breast cancer models revealed that only a small portion of tumor cells is motile¹⁶, and that fast- and slow-migrating cells are present in distinct regions of the tumor^{7,17}. Moreover, additional heterogeneity exists among migrating tumor cells that can, for instance, move randomly or directionally relative to other tumor cells and/or surrounding healthy tissue⁷. However, traditional IVM analyses have considered cancer cells or the tumor microenvironment (TME) as a homogeneous population, thereby ignoring heterogeneity at the single-cell level and the influence of the local microenvironment, where interactions between neighboring cells may affect cellular behavior. To address these complexities, recent studies have used unbiased approaches using morpho-kinetic parameters derived from live *in vitro*¹⁸ or *in vivo* imaging to classify cells and explore the single-cell behavioral landscape as an additional omic layer^{19–21}. In line with this concept of characterizing cellular dynamic properties for cell classification, we have previously developed an analytical platform termed BEHAV3D^{20,22} allowing to perform behavioral phenotyping of engineered T cells targeting cancer. This approach revealed substantial behavioral heterogeneity among the engineered T cells and identified, based on behavioral characteristics, a subpopulation of T cells with increased killing capacity.

However, the implementation of unbiased methods for cellular dynamic analysis, such as BEHAV3D, often demands computational expertise beyond the reach of many biomedical researchers conducting IVM experiments. To democratize the thorough understanding of heterogeneous cancer cell behavior and the microenvironmental components influencing it, there is a need for comprehensive, user-friendly, zero-code analysis tools. Such tools would facilitate unbiased analysis of diverse tumor dynamics, making advanced computational techniques accessible to a broader range of researchers.

Building upon our BEHAV3D pipeline for analyzing engineered T cell behavior²², we introduce BEHAV3D Tumor Profiler (BEHAV3D-TP) (**Fig. 1**), a new zero-code computational framework tailored to map the behavioral patterns of single tumor cells and their interplay with the TME. We

used this novel tool to get more insights into the *in vivo* migration behavior of diffuse midline glioma (DMG), a highly aggressive pediatric brain tumor characterized by invasive growth and its capacity to spread to other regions of the brain and the spinal cord²³. For the first time, by visualizing TME components either during or after IVM, we were able to show that DMG cell migration behavior correlates with specific microenvironmental factors, including tumor-associated macrophages and vasculature. In our study, we specifically focused on cancer cell migration as a specific feature of DMG, but our methodology can easily be extended to other cancer cell behaviors such as intravasation or metastatic colonization. Overall, BEHAV3D-TP provides an accessible computational approach for a better understanding of tumor-TME crosstalk and the functional implications of this communication in heterogeneous tumors.

Results

BEHAV3D Tumor Profiler: an accessible tool for mapping heterogeneity in tumor dynamics

To comprehensively map and analyze the dynamic heterogeneity of tumor cells within their TME we developed BEHAV3D-TP [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler]. Aiming at enhancing accessibility and widespread adoption across scientists studying single tumor dynamics in live imaging, and inspired by previous work on democratizing advanced image analysis^{24–26}, we designed BEHAV3D-TP as a modular Jupyter Notebook featuring an intuitive graphical user interface (GUI) (Figure 1 and Supplementary Video 1). Operating directly within a web browser, the platform only requires a Google account for access. BEHAV3D-TP implementation within the Google Colaboratory (Colab) framework not only ensures efficient cloud-based computation with free and scalable resources but also enhances tool accessibility by eliminating the need for local software installation, a common barrier for inexperienced users. Regardless of programming proficiency, users can effortlessly navigate the computational pipeline to extract comprehensive insights into tumor dynamics (Figure 1). After the time-lapse image file processing and preparation (Figure 1a), BEHAV3D-TP includes a central module called the *heterogeneity module* (Figure 1b), enabling researchers to uncover intricate patterns of tumor behavior. Additionally, we provide two optional modules—the *large-scale phenotyping module* (Figure 1c) and the *small-scale phenotyping module* (Figure 1d)—which, combined with TME visualization during or after IVM, allow for correlating different tumor behavioral patterns with the composition of the TME. To demonstrate the application of BEHAV3D-TP, we utilized single-cell dynamic measurements obtained from intravital imaging of DMG-bearing mice.

Identification of highly heterogeneous *in vivo* cell behavior displayed by DMGs

To get insights into the behavioral heterogeneity among individual DMG cells, we performed IVM on mice implanted with a cranial imaging window to visualize cancer cells expressing H2BmNeonGreen marker for up to 5.4 hours (Figure supplement 1a). We tracked intravitaly imaged DMG cells from different mice and in each imaged position we determined how each DMG cell's position changes relative to the tumor edge (Figure 2a). Next, to untangle the complexity of tumor cell dynamics in an unbiased manner, we used the BEHAV3D-TP *heterogeneity module*, that implements multiparametric single-cell time-series classification allowing to identify distinct single-cell behavioral patterns (Figure 2 a, b). For this, we used kinetic parameters, including displacement, speed, persistence, and movement direction relative to the tumor edge, to measure similitudes between cellular behavior using the dynamic time warping algorithm²², followed by dimensionality reduction to classify cells. The BEHAV3D-TP *heterogeneity module* identified 7 DMG behavioral clusters: *retreating*, *slow retreating*, *erratic*, *static*, *slow*, *slow invading*, and *invading* (Figure 2b, c, d, Supplementary video 2). To accurately describe each of the clusters, we analyzed their most predominant features (Figure 2d) and back-projected the cluster

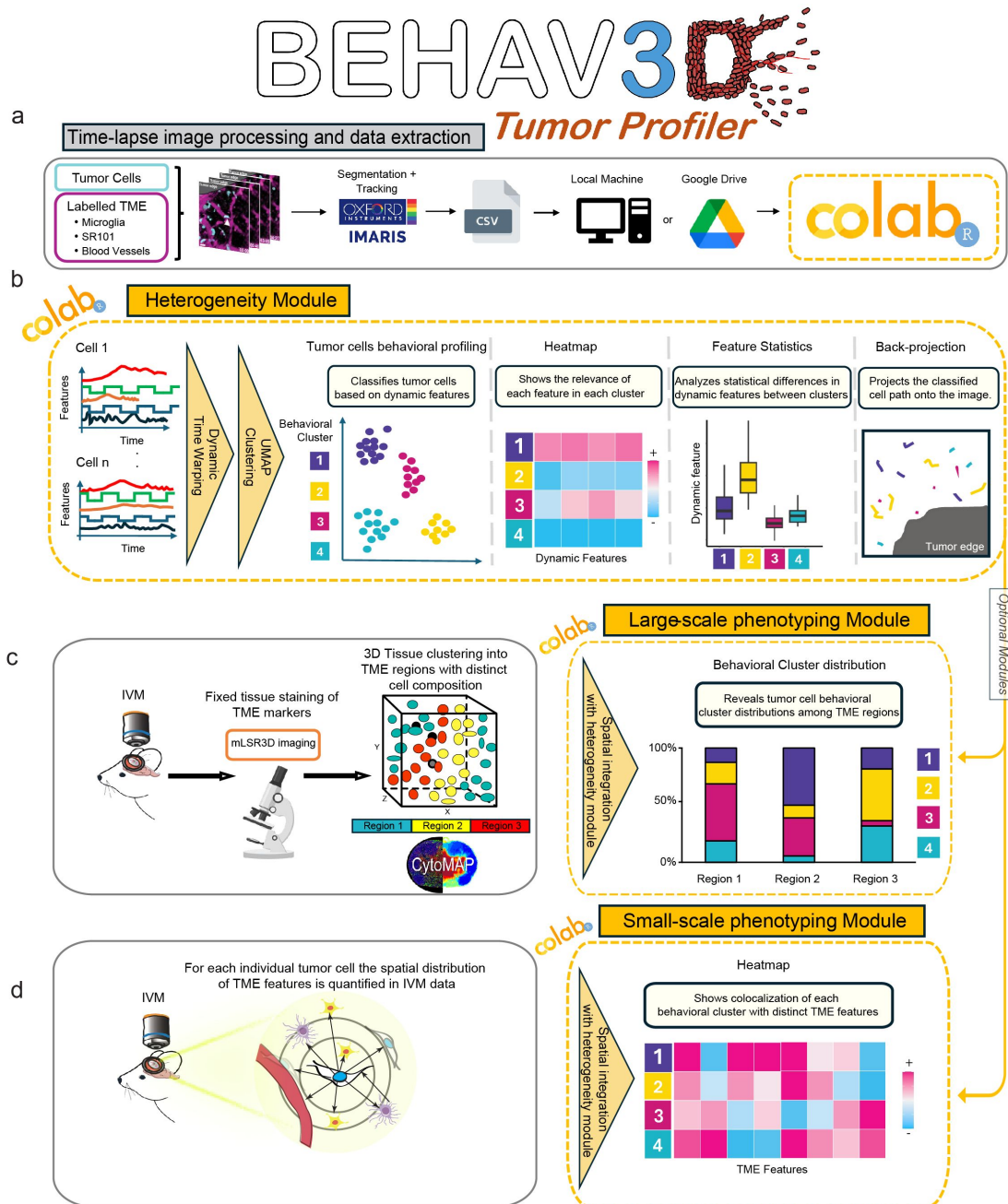


Fig. 1

BEHAV3D Tumor Profiler pipeline.

Schematic representation of the workflow showing data preparation, input and outputs in each of the three modules. The programmatic processing is represented with a yellow background **a**, Time-lapse image processing and preparation, based on segmentation and tracking of tumor and labelled TME cells. Upload to Google Drive is optional but recommended to work in the Google Colab framework. **b**, Heterogeneity Module provides distinct behavioral clusters for DMG cells, and relates them to dynamic features and provides a back-projection of the tracked cells. **c**, **d**, Optional modules, additional to the Heterogeneity module. **c**, Large-scale phenotyping module combines TME large-scale information with DMG behavioral profiling to further depict population frequencies in distinct TME regions. **d**, Small-scale phenotyping module combines TME small-scale information with DMG behavioral profiling to depict how the cellular environment can influence tumor cell behavior.

information into the original time-lapse to visually inspect cell behavior in relation to the tumor (Figure 2a, c, Figure supplement 1b, Supplementary video 2). Of particular interest were cells from the *invading* and *retreating* clusters, characterized by fast movement away from and towards the tumor edge, respectively (Figure 2c-f, and Figure supplement 1c). In fact, in this particular DMG model, approximately 10% of the cancer cells were displaying *invasive* behavior (Figure supplement 1d, e), in line with the intrinsic infiltrative nature of DMG²⁷. Interestingly, about another 10% of DMG cells were moving back to the tumor edge (*retreating*, *slow retreating* clusters) (Figure supplement 1d, e), behavior that has been previously observed for glioblastoma and suggested to be regulated by differential chemotactic signaling by the tumor edge and the surrounding brain parenchyma⁷. We also observed *erratic* cells that display fast non-directed migration behavior, and *static* cells that are non-migratory (Figure 2c-f). Crucially, the distinct behavioral patterns were detected in all intravitaly imaged mice (Figure supplement 1d, e), ruling out any sample bias that might lead to clusters consisting solely of cells from a single tumor or mouse. This underscores the effectiveness of the BEHAV3D-TP in revealing representative heterogeneous tumor behavioral patterns.

Mapping *in vivo* tumor cell migratory properties to distinct TME regions identified through *ex vivo* large-scale 3D imaging

To investigate whether the TME influences behavioral heterogeneity among cancer cells, we developed two complementary approaches to correlate the identified behavioral clusters (Figure 2) to the composition of the TME (Figures 1c, 3a and 4a).

The first approach consists of performing large-scale TME phenotyping and identifying regions with a specific cellular composition and architecture within the TME of intravitaly imaged tumors that have been intravitaly imaged and subsequently fixed. This is accomplished by using multispectral large-scale single-cell resolution 3D (mLSR-3D) imaging data^{28,29} and a spatial analysis framework called CytoMAP³⁰ (Figure 3a). We selected microenvironmental components of interest, such as perivascular niches and glial cells, as they have been previously reported to play an important role in the invasive behavior of glioblastoma cells^{7,31–35}. Specifically, intravitaly imaged brains were collected immediately after the IVM imaging sessions and were fixed and stained for blood vessels (CD31), oligodendrocytes (Olig2), and tumor-associated microglia/macrophages (TAMM) (Iba1) (Figure 3b). Subsequent CytoMAP³⁰ spatial phenotyping analysis identified three different environmental niches: *Void* regions (less vascularized and with low glial infiltration), *TAMM/oligo* regions (enriched in oligodendrocytes, TAMM, and tumor cells), and *TAMM/vascularized* regions (highly vascularized regions enriched in TAMM) (Figure 3b, and Figure supplement 2a-c). To investigate the heterogeneity of tumor cell behavior in relation to the identified microenvironmental niches, BEHAV3D-TP *large-scale phenotyping module* mapped the different niches (*Void*, *TAMM/Oligo*, and *TAMM/vascularized*) identified by CytoMAP onto 3D intravital imaging data (Figures 1c, 3c). In the TME-defined IVM-imaged positions we compared the frequencies of the different behavioral clusters that we identified by with the BEHAV3D-TP *heterogeneity module* (Figure 3d). Interestingly, *invading* cells were more abundant in *TAMM/vascularized* regions compared to *Void* regions that contain a higher proportion of *static* cells, and *TAMM/Oligo* microenvironments contain more *slow retreating* cells compared to the other regions (Figure 3d). Finally, we compared the results obtained with BEHAV3D-TP to a more classical IVM analysis approach that relies on assessing single dynamic parameters. Interestingly, single parameters showed a restricted ability to identify differences in cellular behavior among various environments (Figure supplement 2e-g), underscoring the importance of multiparametric behavioral mapping in revealing more nuanced cellular behavior that cannot be captured solely by individual dynamic parameters.

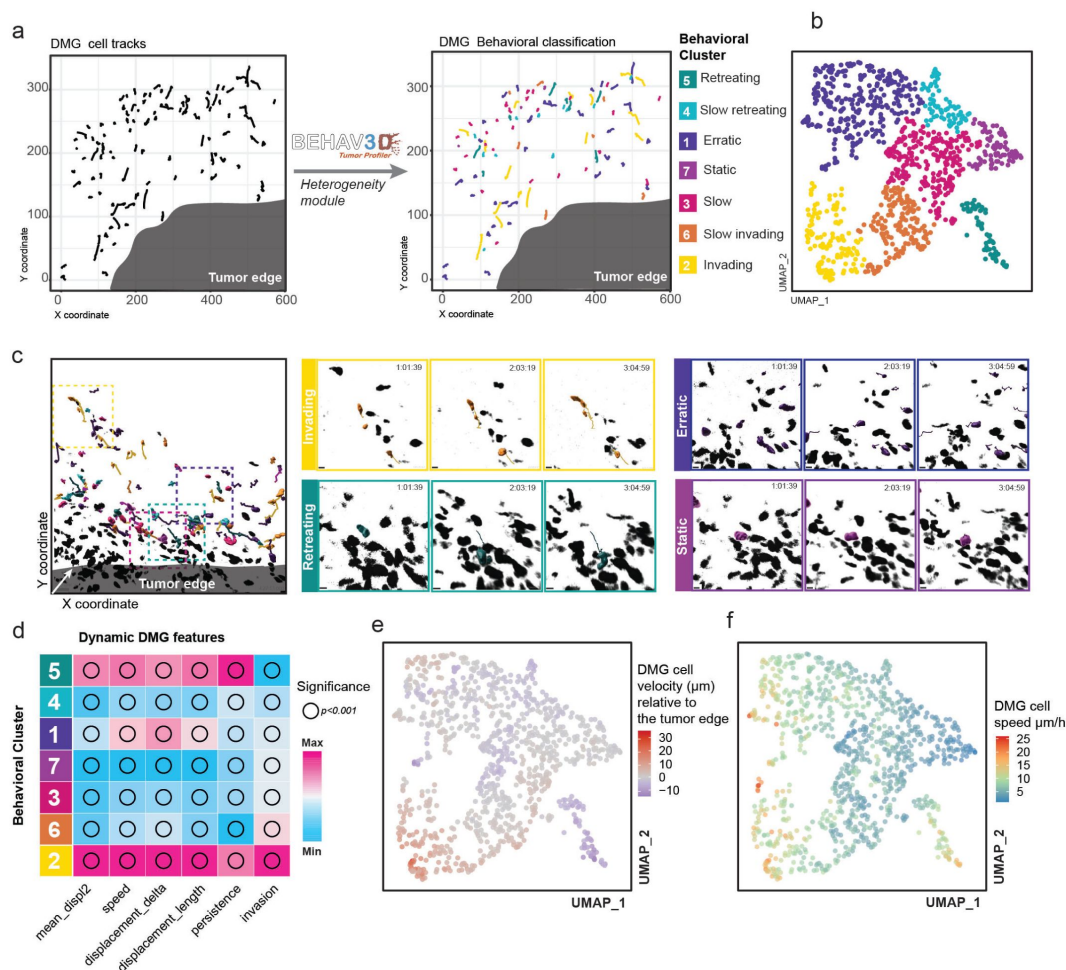


Fig. 2

Behavioral profiling of DMG cells with BEHAV3D-TP heterogeneity module.

a, Representative intravitaly imaged position showing DMG cells tracked relative to tumor Edge (left panel). Tracks were classified according to DMG behavior (right, color coded by cluster). Representative of $n = 18$ independent positions. **b**, UMAP plot showing seven color-coded DMG cell behavioral clusters identified by BEHAV3D-TP. Each datapoint represents one T cell track. **c**, Representative intravitaly imaged positions and enlarged sections for showcasing distinct DMG behavioral clusters *Invading* (yellow), *Retreating* (green), *Erratic* (dark purple) and *Static* (light purple) through the time series. $n = 18$ independent experiments. Scale bars 10µm. **d**, Heatmap depicting relative values of DMG cell features indicated for each cluster, named according to their most distinct characteristics. Arbitrary units in respect to maximal and minimal values for each feature. Bubble size according to significance levels, all features represented have a significance level of $P < 0.001$ (***): *mean_displ2*, *mean squared displacement*; *speed*; *delta displacement*; *displacement length*; *persistence*; *invasion*. P-values represent the significance of differences across clusters tested through ANOVA for each feature. **e**, **f**, UMAP representation of DMG cells' velocity relative to the tumor edge (µm) (**e**) and speed (µm/h) (**f**). Each datapoint represents one T cell track and is colored following a color gradient. For each feature colors represent the mean DMG track values.

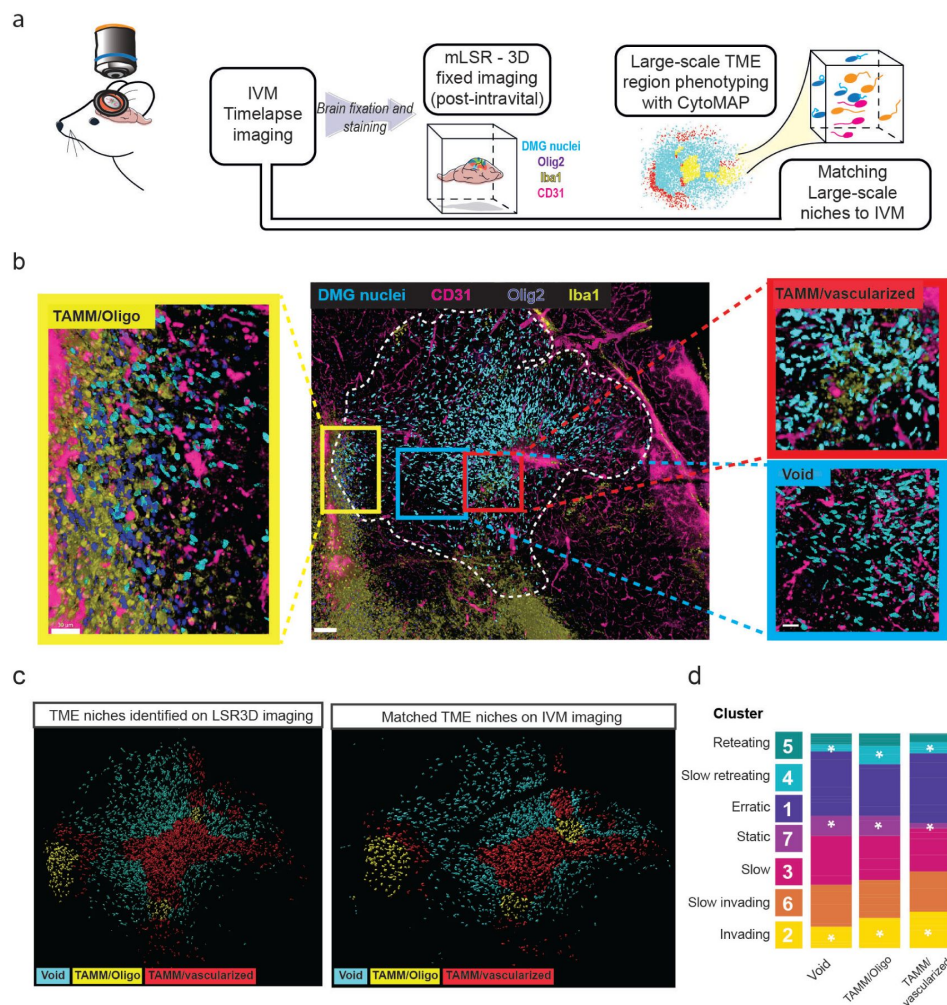


Fig. 3

Mapping distinct DMG behavioral patterns to distinct TME regions.

a, Overview of the large-scale TME region phenotyping workflow. After IVM imaging session, the tissue is fixed and stained and then analyzed using mLSR-3D. CytoMAP Spatial Analysis Toolbox is used to identify distinct large-scale regions, that are subsequently matched to IVM imaging dataset. **b**, Representative fixed multispectral image of a mLSR3D imaged DMG tumor and zoomed-in images of TME large-scale regions: *Void* (blue outline), *TAMM/vascularized* (red outline) and *TAMM/Oligo* (yellow outline). DMG nuclei (blue), CD31 (pink), Olig2 (purple), and Iba1 (yellow) are represented on the imaged. Scale bars, 100 μ m (overview), 30 μ m (zoomed-in regions). **c**, 3D multispectral image DMG cell rendering color-coded per large TME region: *Void* (blue), *TAMM/Oligo* (yellow) and *TAMM/vascularized* (red) (left). Regions were mapped on intravital imaging data (right). **d**, DMG single-cell behavioral cluster frequencies among the TME large-scale regions identified with CytoMAP. *Slow retreating*, $P < 0.05$; *Static*, $P < 0.05$; *Invading*, $P < 0.05$. P-values represent the significance of differences across TME regions tested through ANOVA for each behavioral cluster.

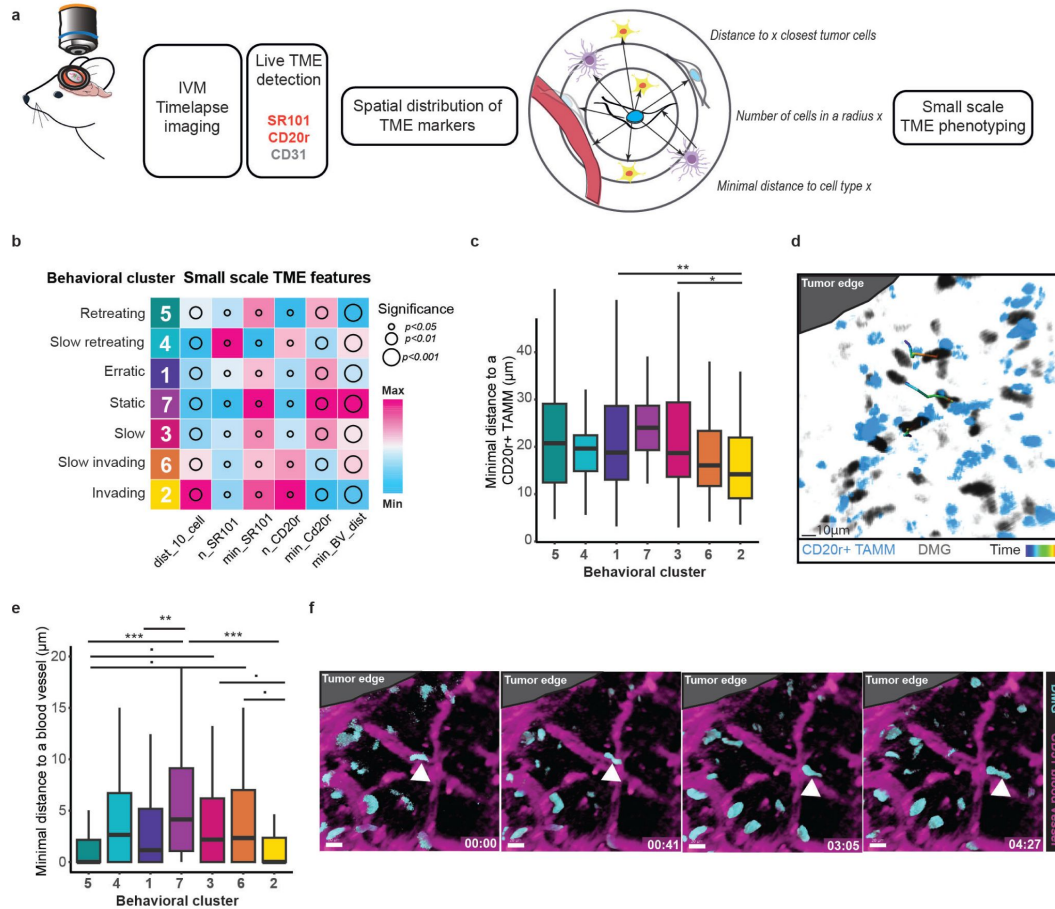


Fig. 4

The BEHAV3D-TP small-scale phenotyping correlates DMG behavioral profiles with microenvironmental composition at a single-cell resolution.

a, Overview of the small-scale TME phenotyping module. During IVM imaging, information about spatial distribution of SR101+ (purple), CD20r+ (purple) and CD31+ (red) cells is collected and processed by the BEHAV3D-TP *small-scale phenotyping module*. Single DMG cell spatial TME features like distance to its neighbors, number of cells in a radius or minimal distance to a cell type were measured. **b**, Heatmap depicting DMG cell small-scale niches' features across distinct DMG behavioral clusters. Arbitrary units in respect to maximal and minimal values for each feature. Bubble size according to significance levels, from smallest ($P < 0.05$ (**)) to medium ($P < 0.01$ (***)) to biggest ($P < 0.001$ (***)): *dist_10_cell*, Distance to nearest 10 cells, $P < 0.01$ (**); *n_SR101*, number of SR101 cells, $P < 0.05$ (*); *min_SR101*, minimal distance to a SR101 cell, $P < 0.05$ (*); *n_CD20r*, number of CD20r cells, $P < 0.05$ (*); *min_CD20r*, minimal distance to a CD20r cell, $P < 0.01$ (**); *min_BV_dist*, minimal distance to a Blood vessel, $P < 0.001$ (***). P-values represent the significance of differences across clusters tested through ANOVA for each feature. **c**, **e**, Boxplot depicting the *Minimal distance to a CD20r+ TAMM* (μm) (**c**) and the *Minimal distance to a blood vessel* (μm) (**e**) across DMG behavioral clusters. *c*, *Invading* versus *Erratic*, $P < 0.01$ (**); *Invading* versus *Slow*, $P < 0.05$ (*). *e*, *Static* versus *Erratic*, $P < 0.01$ (**); *Static* versus *Retreating*, $P < 0.001$ (***); *Static* versus *Invading*, $P < 0.001$ (***); *Slow* versus *Retreating*, $P < 0.1$ (-); *Slow invading* versus *Retreating*, $P < 0.1$ (-); *Slow* versus *Invading*, $P < 0.1$ (-); *Slow invading* versus *Invading*, $P < 0.1$ (-). **d**, Fixed multispectral image showing the movement of DMG cells (grey) relative to CD20r+ TAMM (blue) and the Tumor core. The trajectories show a path from the initial (blue) to the final position (red). Scale bar, 10 μm . **f**, Time lapse multispectral images (left to right) showing the DMG cells (blue) displacement along CD31 blood vessel (purple) away from the tumor edge. Scale bars, 30 μm .

Linkage of *in vivo* tumor cell behavior to the composition of the microenvironment at a single-cell level

With the goal of further refining TME phenotyping to better understand tumor cell behavior determinants, we implemented an alternative approach using the BEHAV3D-TP *small-scale phenotyping module* (Figure 1d, 4a). This module utilizes the detection of TME components during IVM to offer insights into their abundance and spatial relationship with DMG cells at a single-cell level. Together with the *heterogeneity module* that profiles the single-cell behavior of tumor cells (Figure 1c), this additional module allows linking different DMG behavioral profiles with DMG cell small-scale niches (Figures 1d, 4a). For *in vivo* TME labeling, we injected mice with an anti-CD31-AF647 antibody to label blood vessels, and with either SR101, a fluorescent marker known to label oligodendrocytes and astrocytes^{36–39}, or CD20r, a novel molecule described to label microglial cells⁴⁰ (Figure supplement 3a, b). This allowed us to measure spatial TME characteristics of each single DMG cell, such as minimal distance to a certain TME component, the number of these cell types in a certain radius, or the cell density for this particular cell (Figure 1d). Interestingly, the BEHAV3D-TP *small-scale phenotyping module* revealed substantial TME heterogeneity across the different behavioral DMG clusters (Figure 4b). Firstly, compared to *slow* and *slow-retreating* cells, *invading* cells were usually found to be less densely positioned relative to other DMG cells (Figure 4b, and Figure supplement 3c), correlating with their invasive potential into the brain parenchyma. Second, and in line with the data from the large-scale TME phenotyping (Figure 3c), our data suggests that *invading* DMG cells reside in a local environment that is enriched in TAMMs (Figure 4b). Indeed, *invading* cells are closer to TAMMs compared to *erratic* and *slow* migrating cells (Figure 4c), and move directionally towards TAMMs (Figure 4d, Supplementary video 3), suggesting that migration of this behavioral cluster is at least partially mediated by chemotaxis. Third, all fast-migrating behavioral clusters (*invading*, *erratic*, and *retreating*) were found to be generally closer to blood vessels (Figure 4b and e), suggesting that perivascular invasion is a more efficient movement route for these cells (Figure 4f), a mode of migration that has been previously observed for adult glioblastoma cells as well^{7,31,33–35}. Lastly, particularly *slow-retreating* DMG cells were found in regions enriched for SR101⁺ cells (Figure 4b and Figure supplement 3d), perhaps reflecting oligodendrocyte-like (OC-like) tumor DMG cells that have been found near oligodendrocytes⁴¹. Altogether, these data demonstrate the power of our BEHAV3D-TP pipeline in correlating single-cell tumor cell behavior to its local environment, with an important opportunity to identify novel environmental regulators of invasion.

Discussion

Here, we provide a new user-friendly platform tailored for IVM single tumor cell dynamics analysis, termed BEHAV3D-TP, aiming to better understand the mechanisms underlying tumor heterogeneity. In contrast to traditional pipelines that rely on the analysis of single parameters such as cell speed or displacement^{42–44} or that use an artificial threshold to assign a specific behavior to cells¹⁰, BEHAV3D-TP is an unbiased approach based on the integration of multiple parameters measured by IVM. Using the *heterogeneity module* of our BEHAV3D-TP pipeline, we uncovered behavioral phenotypes that have not been described before and that could not have been identified using single parameter analysis. For example, our analysis revealed *invading* and *retreating* clusters, composed of cells with a similar cell speed but with an opposite migration direction, an important biological and clinical difference. Likely, bidirectional migration is mediated by different chemotactic cues and by the composition of the local TME⁴⁵. It is of vital importance for DMG and other cancer patients to better understand the biology underlying the different behavioral phenotypes, especially concerning *invading* cells that are difficult to therapeutically target⁴⁶.

Previous studies suggest a major role for the microenvironment in promoting tumor cell spread into the healthy brain ⁴⁷[47](#)–⁴⁹[49](#). Indeed, using the complementary *large-* and *small-scale TME phenotyping modules* of BEHAV3D-TP, we identified substantial differences in the TME composition of the different behavioral DMG clusters. Our results also show that the small-scale TME analysis has a higher predictive value compared to the large-scale TME analysis, suggesting that a cell's direct microenvironment is a more important regulator of its behavior compared to its macroenvironment. Future experiments directed towards identifying more TME components, for example using multispectral LSR-3D imaging²⁸[28](#), could therefore further refine the *large-scale phenotyping module* by detecting subregion heterogeneity.

IVM imaging allows for simultaneous labeling of various TME components and can be used to study live their association with tumor cell behavior. Indeed, we found that TAMMs and blood vessels were associated with specific behavioral phenotypes, showing the power of BEHAV3D-TP *small-scale phenotyping module* in identifying novel regulators of cellular behavior. This is exemplified by our finding that *slow-retreating* DMG cells were found in environments enriched for SR101⁺ cells, an observation that has not been made before. To take further advantage of BEHAV3D-TP, future work should be directed toward identifying more TME components, including cells of the adaptive immune system. For example, it would be interesting to use our pipeline on recently developed immunocompetent mouse models of DMG ⁵⁰[50](#),⁵¹[51](#), or in mice with a humanized immune system⁵²[52](#).

To enhance the accessibility of BEHAV3D-TP for researchers keen on studying tumor cell behavioral profiles through live imaging, we have incorporated our software in a Google Colab notebook, allowing researchers with minimal programming skills to employ the tool online without the requirement to install specific software environments nor the need to directly interact with the code. The user can easily install the necessary dependencies, set up the working environment, process datasets and adjust parameters without requiring deep coding skills. Google Colab provides the necessary computational resources to be able to run the pipeline without demanding a powerful local machine. However, there are RAM usage and runtime limitations that could affect the analysis, given a big enough dataset in the free version of Google Colab. Given this and data privacy concerns, the workflow can also be run locally via a Jupyter Notebook, that can be downloaded from the Google Colab notebook. Aligned with the goals of BEHAV3D-TP, other methods utilizing the Google Colab framework are emerging, such as CellTracksColab²⁴[24](#), which is a general tool for tracking data analysis. While CellTracksColab offers valuable insights into tracking data, BEHAV3D-TP is specifically designed for IVM data and includes additional features. These features include directionality metrics crucial for invasion studies, detailed analysis of identified behavioral populations, data back-projection and correlation of these behaviors with both large-scale and small-scale TME features. This comprehensive approach provides a deeper understanding of the TME drivers behind intratumoral heterogeneity, making BEHAV3D-TP a robust tool for researchers investigating the complex behaviors of tumor cells in their natural environments.

In summary, BEHAV3D-TP allows to get insights into the heterogeneity of tumor cell migration behavior in an accessible way; however, it would be exciting to extend the application to other behaviors, such as therapy response to study TME-mediated drug resistance. Such an endeavor may benefit from including additional parameters beyond kinetic features. For example, others have additionally used morphological parameters ¹⁹[19](#),²¹[21](#), and we envision that a multiparametric analysis based on a combination of morpho-kinetic features may further refine behavioral phenotyping.

Methods

Animal material

NOD.Cg-PrkdcscidIl2rgtm1Wjl/SzJ (NSG) mice were purchased from Charles River Laboratories (France). Experiments were conducted per Institutional Guidelines under acquired permission from the local Ethical Committee and as per current Dutch laws on Animal Experimentation. Mice were housed under sterile conditions using an individually ventilated cage (IVC) system and were fed with sterile food and water.

Human DMG sphere-forming culture

DMG007 (HSJD-DIPG-007) cells established from primary DMG material were grown in a base medium (TSM) consisting of 48% DMEM/F12 and 48% Neurobasal medium (Thermo Fisher) supplemented with GlutaMAX, 100mM Sodium Pyruvate, MEM non-essential amino acids and 1M Hepes buffer, at 1% each. Primocin (InvivoGen) at 50mg/ml was additionally added to the TSM. Working medium was prepared by adding 20ng/ml of human EGF and human basic FGF, 10ng/ml of PDGF-AA and PDGF-BB (Peprotech) and 2ug/ml of Heparin (StemCell Technologies). DMG cells were cultured at 37 °C in a humidified atmosphere with 5% CO₂.

Fluorescent reporter cloning and lentiviral transduction

Fluorescent reporter construct was based on the combination of Histone2B-mNeonGreen, P2A and mScarlet I-CAAX gifted from Dr. Hugo Snippert (UMC Utrecht, The Netherlands), and inserted by In-Fusion HD Cloning Plus (TakaraBio) into a lentiviral vector including cPPT/CTS, WPRE, an EF1a promoter and IRES-puromycin-resistance cassette.¹¹ Subsequently the above-mentioned elements were amplified by PCR and the amplicon was assembled in the final lentiviral backbone using Gibson Assembly (NEB). Sequences were validated using Sanger sequencing. DMG007 cells were transduced using a standard lentiviral transduction protocol and selected using puromycin to achieve a stable cell line.

Cranial imaging window (CIW) surgery and tumor cell injection

CIW surgery and tumor cell injection were performed on the same day as previously described^{7,10}. Briefly, mice were sedated with Hypnorm (Fluanison [neuroleptic] + Fentanyl [opioid]) (0.4 ml/kg) + Midazolam [benzodiazepine sedative] (2 mg/kg) at a dose of 1:1:2 in sterile water and mounted in a stereotactic frame. The head was secured using a nose clamp and two custom-made 3D printed ear bars, made out of Polylactic Acid (PLA, Geeetech, 700-001-043). The head was shaved, and the skin was cut circularly. The mouse was placed under a stereo-microscope to ensure precise surgical manipulation. The periosteum was scraped, and a circular groove of 5 mm diameter was drilled over the right parietal bone. The bone flap was lifted under a drop of cortex buffer (125 mM NaCl, 5 mM KCl, 10 mM glucose, 10 mM HEPES buffer, 2 mM MgSO₄, and 2 mM CaCl₂, pH 7.4), and the dura mater was removed. Gelfoam sponge (Pfizer) was used to stop potential bleedings. Next, 1 × 10⁵ DMG007-H2B-mNeonGreen cells resuspended in 3 µl phosphate-buffered saline (PBS) were injected stereotactically using a 5 µl Hamilton syringe with a 2 pt style in the middle of the craniotomy at a depth of 0.5 mm. The exposed brain was sealed with silicone oil and a 6 mm coverslip was glued on top. Dental acrylic cement (Vertex) was applied on the skull surface, covering the edge of the coverslip, and a custom-made 3D printed ring made of biocompatible Polylactide acid (PLA, Geeetech, 700-001-0433) was glued around the coverslip to provide fixation during intravital imaging and to serve as a reservoir for a water drop required for the objective of the microscope. A single dose of 100 µg/kg of buprenorphine (Temgesic, BD pharmaceutical limited) was administered for post-surgical pain relief. Mice were closely monitored twice per week to assess behavior, reactivity, and appearance.

Intravital imaging

Mice were intravitaly imaged as previously described [7](#),[10](#). In short, were sedated with isoflurane and placed face-down in a stereotactic frame. The Polylactide ring was used to fix the mouse's head to the frame with custom-made Polylactide bars. Time-lapse images of the entire tumor volume were acquired for a maximum of 5.4 hours. The minimal time interval between serial images was set to 20 minutes. For tile scans, images of the complete z-stack of the tumor were acquired, with a step size of 2 μm . In a group of 3 mice, oligodendrocytes and astrocytes were imaged by intravenous injection of 50 μL SR101 (ThermoFisher) at a concentration of 5 mM dissolved in PBS. In a different group of 3 mice, TAMMs were imaged by intravenous injection of 50 μL 100 μM CDr20 [40](#). Simultaneously with the injection of either SR101 or CDr20, all mice received an intravenous injection of 10 μL CD31-AF647 antibody (ThermoFisher, A14716) to visualize the blood vessels.

Intravital imaging was performed on an upright FVMPE-RS multiphoton microscope (Olympus) equipped with a MaiTai DeepSee and an InSight DeepSee laser (Spectra-Physics), a 25x/1.05 numerical aperture water objective, and two GaAsP and two Multialkali photomultiplier tubes (Olympus). Images were acquired with simultaneous imaging with the MaiTai (960 nm) and InSight (1100 nm) lasers. Green, red, and far-red fluorescence was separated by 570 nm and 650 nm dichroic mirrors, and 610/70 nm and 705/90 nm filters (Semrock) and collected by GaAsP or Multialkali photomultiplier tubes. Scanning was performed in a bidirectional mode with a resonant scanner at 400 Hz and 12 bit, with a zoom of 1x, 512 $\times$ 512 pixels. For optimal detection of blood vessels (CD31-AF647), a single tile scan was made at the last timepoint with InSight laser tuned at 1250 nm. At the postprocessing step, this image was co-registered to the time-lapse movie.

Co-registration of IVM imaged blood vessel imaging

CD31-AF647 tile scan featuring optimal visualization of blood vessels during live imaging was co-registered to the last timepoint of the time-lapse movie. Both images (last timepoint of time-lapse tile-scan and blood vessel tile-scan) were visually inspected to determine and annotate the internal landmarks using Imaris (Oxford Instruments), versions 9.5 to 9.6. These landmarks served as a reference for the co-registration software to overlap both images. To perform the co-registration, a modification of *elastix* [53](#) was used; a python-based open-source software based on Insight Segmentation and Registration Toolkit (ITK) plus SimpleElastix [54](#): a collection of different algorithms used on medical image registration. Due to the extensive number of potential parameter combinations, specific parameters were manually chosen. This selection was conducted using the Slicer3D visualization software, version 5.2.2.

Immunohistochemistry

3D immunohistochemistry was performed as described before [29](#). After the live imaging session, mouse brains were immersed in 5 ml 4% paraformaldehyde (PFA) pH 7.4 overnight on ice. After fixation, brains were blocked in 5-10 ml Wash Buffer 1 (2 ml Tween-20, 2 ml Triton X-100, 2 ml 10% SDS, and 2 g BSA in 1 l PBS) for 5 hours. For multiplex immunolabeling, a two- or three-round staining protocol was used as previously described [28](#),[29](#). Washing and incubation steps were performed in Wash Buffer 2 (1 ml Triton X-100, 2 ml 10% SDS, and 2 g BSA in 1 l PBS). A thick (1 mm) slice of the cortex part that was under the cranial imaging window was dissected with a scalpel before proceeding to immunolabeling. The following primary antibodies were used: rabbit anti-Olig2 1:100 (AB9610, Merk), goat anti-Iba1 1:200 (NB100-1028SS, NovusBio) and CD31-AF647 1:250 (A14716, ThermoFisher). As secondary antibodies, donkey anti-rabbit AF405 1:250 (ab175651, Abcam) and donkey anti-goat AF633 1:250 (A21082, ThermoFisher) were used. After the last washing step, tissues were optically cleared by three stepwise incubations of an increasing concentration of FUnGI [55](#) clearing agent (25%/50%/75%, diluted in PBS) for 1h at room temperature [28](#). The final incubation step with 100% FUnGI was performed overnight at 4°C.

mLSR3D imaging

mLSR3D imaging^{28,29} of thick slices of fixed mouse cortex was performed using a Zeiss LSM880 system equipped with a 32-channel spectral detector. The signal from all fluorophores present in the samples was collected in a single acquisition and linear unmixing was used to obtain separate channels. Unmixing was carried out by the Online Fingerprinting mode of the microscope. Pre-acquired references of each single fluorophore were used to enable multispectral on-the-fly unmixing. Imaging was performed using a 25x multi-immersion objective 0.8 NA (Zeiss 420852-9871) with a working distance of 570µm. Tile scans were acquired using a voxel size of 0.332x0.332x1.200 µm, a pixel dwell of 2µs, and a 10% tile overlap.

Image processing

For 3D visualization, shift correction, object rendering, and cell tracking, time-lapse movies were processed with Imaris (Oxford Instruments), versions 9.5 to 10.1.

Cell tracking

The *Channel Arithmetics* Xtension was used for channel unmixing of overlapping signals. The Spots ImarisTrack module was used for object detection and semi-automated tracking of tumor cells (autoregressive motion). Tumor cell tracks were manually corrected to ensure accurate tracking. Around 50-100 cells per position were manually corrected and labeled as such for posterior selection of accurate tracks. To determine the direction of tumor cell movement relative to the tumor core, at the image edge closer to the tumor core a “Tumor edge” surface object was manually rendered using the contour tool. The *Distance Transformation* Xtension was used to measure the distance between tumor cells and the “Tumor edge”. For tracked tumor cells, time-lapse data containing the coordinates of each cell, the values of cell speed, mean square displacement, displacement delta length, displacement length, and distance to “Tumor edge” were exported. See <https://qbi.uq.edu.au/files/40820/ImarisManual.pdf> for statistical details.

Microenvironmental factors rendering

For rendering the microenvironmental factors detected during IVM or by post-IVM LSR-3D imaging, the Surface and Spots modules of Imaris were used. Surfaces were used for the SR101, CDr20, Olig2, DAPI, and Iba1 stainings, and Spots were used for CD31+ protruding or elongated structures, as previously described³⁰. For rendered structures, positional data containing the coordinates of objects were exported.

Tumor large-scale spatial phenotyping with Cytomap

LSR-3D imaging data was processed for large-scale spatial phenotyping using the Cytomap Spatial Analysis Toolbox³⁰. Briefly, neighborhoods with a 100 µm radius were generated. Neighborhoods were then clustered into 2 regions based on DMG cell positioning using the David Balwin algorithm. Using the gating tool, the tumor regions (containing DMG objects) were defined. Next, the neighborhoods were clustered again into 3 different regions based on the distribution of Iba1, Olig2, and CD31: *Void*, *TAMM/Oligo*, and *TAMM/vascularized*. By using the *Manually Defined Regions* option, only the tumor region was classified. Finally, for each region, fold-change values were calculated and exported using the *Region Statistics* tab. A 3D image of the cluster distribution was exported for each sample (Figure supplement 2a) and then used to visually assign phenotypical regions in the IVM movies (**Figure 3c**). For each mouse, one or two positions for each type of region were cropped and analyzed for tumor cell behavior as described above (*Cell tracking*).

BEHAV3D Tumor Profiler framework

BEHAV3D-TP was developed using the R Studio version 4.3.3. It features three modules: 1) *Heterogeneity module* with optional 2) *Large-scale and/or* 3) *Small-scale phenotyping modules*. All of the modules are included in the same script available in Github [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler], but can be run independently from one another. To acquire this independency, the optional modules are directly combined to the necessary sections of the *Heterogeneity module*. Consequently, the user can use whichever module is best suited for their purposes. In order to further facilitate public access to the BEHAV3D-TP, a Google Colab notebook has also been created accessible through the same Github page [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler]. This environment is friendly to anyone regardless of their coding expertise as it has a step-by-step follow-through execution through the pipeline. Nevertheless, a Wiki demo run and a video tutorial (Supplementary Video 1) are also provided through the GitHub page [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler] for further clarification, if necessary. In this Colab notebook, the user can easily import data, modify analysis parameters, execute the pipeline and export the obtained results without any computational power demands or computational knowledge. Additionally, the original R version and R Markdown demo scripts are also provided.

Heterogeneity module

To account for missing time points and to create a time series with the same time interval for each tumor cell time series, linear interpolation was used to estimate the values of missing time points. To compare time series independently, tracks were cropped to the minimal common length among all tracks. In our case, to a 2.6 hour duration. For tracked tumor cells, time-lapse data containing the coordinates of each cell, the values of cell speed, mean square displacement, displacement delta length, displacement length, and relative distance to “Tumor edge” were used to calculate a cross-distance matrix between time series. First, we performed a principal component (PC) analysis and selected the number of PCs that explained at least 90% of data variance (3).

The previously calculated PCs were then used to compute the cross-distance matrix between each tumor cell multivariate time series using the dynamic time warping algorithm from the package “dtwclust”. To visualize heterogeneous tumor cell behaviors in 2 dimensions, dimensionality reduction on the cross-distance matrix was performed by the Uniform Manifold Approximation and Projection method (“umap” package). Clustering was performed using the k-means clustering algorithm. For each cluster, the average values of distinct behavioral and environmental components were calculated and plotted as a heatmap (“pheatmap”).

Large-scale phenotyping module

The information obtained from mLSR3D and Cytomap about large-scale TME regions were combined with the DMG cells behavioral information to examine the large-scale features of tumor cells. For tumor cells, the mean speed, square displacement and raw movement were analyzed.

Small-scale phenotyping module

The positional coordinates of tracked tumor cells and detected microenvironmental factors were used to determine the small-scale landscape of tumor cells. For tumor cells, the average distance to the closest 10 neighbors was measured to determine tumor cell density. For each tumor cell and each environmental component, the number of microenvironmental objects in a 30- μ m radius and the distance to the closest object were computed. For each behavioral cluster, the values of closest neighbors, minimal distance to each environmental factor, and number of environmental objects in the 30 μ m radius were plotted and compared.

Statistical analyses

Statistical analyses were performed using R. Results are represented as mean $\pm$ s.e.m. unless indicated otherwise; n represents independent positions in 6 biological replicates. Two-tailed unpaired t-tests were performed between two groups unless indicated otherwise. To compare frequencies of different behavioral signatures between large scale regions identified, a Tukey Honestly Significant Difference (HSD) adjusted Estimated marginal means (emmeans) pairwise comparison test was applied. To compare the distribution of behavioral clusters between the environmental clusters, a mixed linear regression followed by a one-way ANOVA was performed. To compare the distribution of different behavioral and environmental features between the behavioral clusters, a one-way ANOVA followed by Tukey HSD correction was performed.

Data availability

Data will be made publicly available after publication.

Code availability

We provide the BEHAV3D-TP framework in a Google Colab user-friendly environment [<https://colab.research.google.com/drive/1JI7ysqFf3tvd6Df4YUsSZ8RbuXw8wba?usp=sharing>], as well as in an R script format in Github [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler]. Additionally, a Wiki [https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler/wiki] to follow through a demo dataset as well as a video tutorial (Supplementary Video 1) has been developed for further assistance.

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Reviewer #1 (Public review):

Summary:

Intravital microscopy (IVM) is a powerful tool that facilitates live imaging of individual cells over time in vivo in their native 3D tissue environment. Extracting and analysing multi-parametric data from IVM images however is challenging, particularly for researchers with limited programming and image analysis skills. In this work, Rios-Jimenez and Zomer et al have developed a 'zero-code' accessible computational framework (BEHAV3D-Tumour Profiler) designed to facilitate unbiased analysis of IVM data to investigate tumour cell dynamics (via the tool's central 'heterogeneity module') and their interactions with the tumour microenvironment (via the 'large-scale phenotyping' and 'small-scale phenotyping' modules). It is designed as an open-source modular Jupyter Notebook with a user-friendly graphical user interface and can be implemented with Google Colab, facilitating efficient, cloud-based computational analysis at no cost.

To demonstrate the utility of BEHAV3D-TP, they apply the pipeline to timelapse IVM imaging datasets to investigate the in vivo migratory behaviour of fluorescently labelled DMG cells in tumour-bearing mice. Using the tool's 'heterogeneity module' they were able to identify distinct single-cell behavioural patterns (based on multiple parameters such as directionality, speed, displacement, and distance from tumour edge) which was used to group cells into distinct categories (e.g. retreating, invasive, static, erratic). They next applied the framework's 'large-scale phenotyping' and 'small-scale phenotyping' modules to investigate whether the tumour microenvironment (TME) may influence the distinct migratory behaviours identified. To achieve this, they combine TME visualisation in vivo during IVM (using fluorescent probes to label distinct TME components) or ex vivo after IVM (by large-scale imaging of harvested, immunostained tumours) to correlate different tumour behavioural patterns with the composition of the TME. They conclude that this tool has helped reveal links between TME composition (e.g. degree of vascularisation, presence of tumour-associated macrophages) and the invasiveness and directionality of tumour cells, which would have been challenging to identify when analysing single kinetic parameters in isolation.

A key limitation of the pipeline is that it does not overcome the main challenges and bottlenecks associated with processing and extracting quantitative cellular data from timelapse and longitudinal intravital images. This includes correcting breathing-induced movement artifacts, automated registration of longitudinal images taken over days/weeks, and accurate, automated segmentation and tracking of individual cells over time. Indeed, there are currently no standardised computational methods available for IVM data processing and analysis, with most laboratories relying on custom-built solutions or manual methods. This isn't made explicit in the manuscript early on (described below), and the researchers rely on expensive software packages such as IMARIS for image processing and data extraction to feed the required parameters into their pipeline. This limitation unfortunately reduces the likely impact of BEHAV3D-TP on the IVM field.

Nonetheless, this computational framework appears to represent a useful and comparatively user-friendly tool to analyse dynamic multi-parametric data to help identify patterns in cell migratory behaviours, and to assess whether these behaviours might be influenced by neighbouring cells and structures in their microenvironment. When combined with other methods, it, therefore, has the potential to be a valuable addition to a researcher's IVM analysis 'tool-box'.

Strengths:

- (1) The figures are clearly presented, and the manuscript is easy to follow.
- (2) The pipeline appears to be intuitive and user-friendly for researchers with limited computational expertise. A detailed step-by-step video is also included to support its uptake.
- (3) The different computational modules have been tested using a relevant dataset.
- (4) All code is open source, and the pipeline can be implemented with Google Colab.
- (5) The tool combines multiple dynamic parameters extracted from time-lapse IVM images to identify single-cell behavioural patterns and to cluster cells into distinct groups sharing similar behaviours, and provides avenues to map these onto in vivo or ex vivo imaging data of the tumour microenvironment.

Weaknesses:

- (1) As highlighted above, the tool does not facilitate the extraction of quantitative kinetic cellular parameters (e.g. speed, directionality, persistence, and displacement) from intravital images. Indeed, to use the tool researchers must first extract dynamic cellular parameters

from their IVM datasets, requiring access to expensive software (e.g. IMARIS as used here) and/or above-average computational expertise to develop and use custom-made open-source solutions. This limitation is not made explicit or discussed in the text.

(2) The number of cells (e.g. per behavioural cluster), and the number of independent mice, represented in each result figure, is not included in the figure legends and are difficult to ascertain from the methods.

(3) The data used to test the pipeline in this manuscript is currently not available, making it difficult to assess its usability. It would be important to include this for researchers to use as a 'training dataset'.

(4) Precisely how the BEHAV3D-TP large-scale phenotyping module can map large-scale spatial phenotyping data generated using LSR-3D imaging data and Cytomap to 3D intravital imaging movies is unclear. Further details in the text and methods would be beneficial to aid understanding.

(5) The analysis provides only preliminary evidence in support of the authors' conclusions on DMG cell migratory behaviours and their relationship with components of the tumour microenvironment. Conclusions should therefore be tempered in the absence of additional experiments and controls.

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Reviewer #2 (Public review):

Summary:

The authors produce a new tool, BEHAV3D to analyse tracking data and to integrate these analyses with large and small-scale architectural features of the tissue. This is similar to several other published methods to analyse spatiotemporal data, however, the connection to tissue features is a nice addition, as is the lack of requirement for coding. The tool is then used to analyse tracking data of tumour cells in diffuse midline glioma. They suggest that 7 clusters exist within these tracks and that they differ spatially. They ultimately suggest that these behaviours occur in distinct spatial areas as determined by CytoMAP.

Strengths:

(1) The tool appears relatively user-friendly and is open source. The combination with CytoMAP represents a nice option for researchers.

- The identification of associations between cell track phenotype and spatial features is exciting and the diffuse midline glioma data nicely demonstrates how this could be used.

Weaknesses:

(1) The strength of democratizing this kind of analysis is undercut by the reliance upon Imaris for segmentation, so it would be nice if this was changed to an open-source option for track generation.

(2) The main issue is with the interpretation of the biological data in Figure 3 where ANOVA was used to analyse the proportional distribution of different clusters. Firstly the *n* is not listed so it is unclear if this represents an *n* of 3 where each mouse is an individual or whether each track is being treated as a test unit. If the latter this is seriously flawed as these tracks can't be treated as independent. Also, a more appropriate test would be something like a Chi-squared test or Fisher's exact test. Also, no error bars are included on the stacked bar graphs making interpretation impossible. Ultimately this is severely flawed and also appears

to show very small differences which may be statistically different but may not represent biologically important findings. This would need further study.

(3) Figure 4 has similar statistical issues in that the *n* is not listed and, again, it is unclear whether they are treating each cell track as independent which, again, would be inappropriate. The best practice for this type of data would be the use of super plots as outlined in Lord et al. (2020) JCI - SuperPlots: Communicating reproducibility and variability in cell biology.

(4) The main issue that this raises is that the large-scale phenotyping module and the heterogeneity module appear designed to produce these statistical analyses that are used in these figures and, if they are based on the assumption that each track is independent, then this will produce inappropriate analyses as a default.

<https://doi.org/10.7554/eLife.102097.1.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Rios-Jimenez developed a computational tool, BEHAV3D Tumor Profiler, to analyze intravital imaging data and extract distinctive tumor cell migratory phenotypes based on the quantified 3D image data.

Weaknesses:

(1) The most challenging task of analyzing 3D time-lapse imaging data is to accurately segment and track the individual cells in 3D over a long time duration. BEHAV3D Tumor Profiler did not provide any new advancement in this regard, and instead relies on commercial software, Imaris, for this critical step. Imaris is known to have a very high error rate when used for analyzing 3D time-lapse data. In the Methods section, the authors themselves stated that "Tumor cell tracks were manually corrected to ensure accurate tracking". Based on our own experience of using Imaris, such manual correction is tedious and often required for every time step of the movie. Therefore, Imaris is not a satisfactory tool for analyzing 3D time-lapse data. Moreover, Imaris is expensive and many research labs probably can't afford to buy it. The fact that BEHAV3D Tumor Profiler critically depends on the faulty ImarisTrack module makes it unclear whether the BEHAV3D tool or the results are reliable.

(2) The authors developed a "Heterogeneity module" to extract distinctive tumor migratory phenotypes from the cell tracks quantified by Imaris. The cell tracks of the individual tumor cells are all quite short, indicating relatively low motility of the tumor cells. It's unclear whether such short migratory tracks are sufficient to warrant the PCA analysis to identify the 7 distinctive migratory phenotypes shown in Figure 2d. It's also unclear whether these 7 migratory phenotypes correspond to unique functional phenotypes.

(3) Using only motility to classify tumor cell behaviours in the tumor microenvironment (TME) is probably not sufficient to capture the tumor cell difference. There are also other non-tumor cell types in the TME. If the authors aim to develop a computational tool that can elucidate tumor cell behaviors in the TME, they should consider other tumor cell features, e.g., morphology, proliferation state, and tumor cell interaction with other cell types, e.g., fibroblasts and distinct immune cells.

(4) The authors have already published two papers on BEHAV3D [Alieva M et al. Nat Protoc. 2024 Jul;19(7): 2052-2084; Dekkers JF, et al. Nat Biotechnol. 2023 Jan;41(1):60-69]. Although the previous two papers used BEHAV3D to analyze T cells, the basic pipeline and computational

steps are similar, in particular regarding cell segmentation and tracking. The addition of a "Heterogeneity module" based on PCA analysis does not make a significant advancement in terms of image analysis and quantification.

<https://doi.org/10.7554/eLife.102097.1.sa1>

Author response:

We want to thank the reviewers for their positive and constructive comments on the manuscript. We already addressed some of their concerns and are planning the following revisions to both BEHAV3D-TP and the corresponding manuscript to address the reviewers' comments. Below, we provide a response to the most significant comments, followed by a detailed, point-by-point response:

(1) We acknowledge the reviewer's suggestion to incorporate open-source segmentation and tracking functionalities, increasing its accessibility to a wider user base; however, these additions fall outside the primary scope of our current work and represent a substantial undertaking in their own right. This topic has been comprehensively explored in other studies (e.g. <https://doi.org/10.4049/jimmunol.2100811> ; <https://doi.org/10.7554/eLife.60547> ; <https://doi.org/10.1016/j.media.2022.102358> ; <https://doi.org/10.1038/s41592-024-02295-6>), which we will cite in our revised manuscript as indicated in our responses to the reviewers' comments. Instead, the goal of our manuscript is to provide an analytical framework for processing data generated by existing segmentation and tracking pipelines. In our analyses, we used data processed with Imaris, a commercial software that, despite its limitations, is widely used by the intravital microscopy community due to its user-friendly platform for 3D image visualization and analysis. Nevertheless, to enhance compatibility with tracking data from various pipelines, we have modified our tool to accept data formats, such as those generated by open-source Fiji plugins like TrackMate (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). These updates are available in our GitHub repository, and we will describe this feature in the revised manuscript to emphasize compatibility with segmented and tracked data from diverse open-source platforms.

(2) We appreciate the reviewer's suggestion to incorporate additional features into our analytical pipeline. In response, we have already updated the GitHub repository to allow users to input and select which features (dynamic, morphological, or spatial) they wish to include in the analysis (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#feature-selection). In the revised manuscript, we will highlight this new functionality and provide examples using alternative datasets to demonstrate the application of these features.

(3) We appreciate the constructive feedback of reviewers #1 and #2 regarding the statistical analysis and interpretation of the data presented in Figures 3 and 4. We understand the importance of clarity and rigor in data analysis and presentation, and we are committed to addressing the concerns raised in the revised version of the manuscript.

(4) We appreciate Reviewer #1's suggestion regarding the inclusion of demo data, as we believe it would greatly enhance the usability of our pipeline. We acknowledge that this was an oversight on our part. To address this, we have now added demo data to our GitHub repository (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler/tree/BEHAV3D_TP-v2.0/demo_datasets). In the upcoming revised manuscript, we will also ensure to reference this addition. Additionally, we will provide both original and processed IVM movie samples to support users in navigating the complete pipeline effectively.

(5) Finally, we agree with the reviewers to make some small changes to the manuscript based on their feedback.

Below we provide a point-by-point response to the reviewers' comments, along with proposed revisions.

Reviewer #1:

Comment: A key limitation of the pipeline is that it does not overcome the main challenges and bottlenecks associated with processing and extracting quantitative cellular data from timelapse and longitudinal intravital images. This includes correcting breathing-induced movement artifacts, automated registration of longitudinal images taken over days/weeks, and accurate, automated segmentation and tracking of individual cells over time. Indeed, there are currently no standardised computational methods available for IVM data processing and analysis, with most laboratories relying on custom-built solutions or manual methods. This isn't made explicit in the manuscript early on (described below), and the researchers rely on expensive software packages such as IMARIS for image processing and data extraction to feed the required parameters into their pipeline. This limitation unfortunately reduces the likely impact of BEHAV3D-TP on the IVM field.

As highlighted above, the tool does not facilitate the extraction of quantitative kinetic cellular parameters (e.g. speed, directionality, persistence, and displacement) from intravital images. Indeed, to use the tool researchers must first extract dynamic cellular parameters from their IVM datasets, requiring access to expensive software (e.g. IMARIS as used here) and/or above-average computational expertise to develop and use custom-made open-source solutions. This limitation is not made explicit or discussed in the text.

As mentioned previously, we agree with the reviewer that image processing steps, such as segmentation, tracking, and motion correction, present significant challenges in intravital microscopy (IVM) data processing. While these aspects are being addressed by other researchers, our publication centers on the analysis of acquired data rather than on the image processing itself. Our motivation, as outlined in the manuscript, arises from our own experience: despite the substantial effort invested in image processing, researchers often rely on simplistic analytical approaches, such as averaging single parameters and comparing them across conditions. These approaches tend to overlook potential tumor heterogeneity.

Our work aimed to develop an analytical tool that provides a comprehensive framework for extracting more insights from processed IVM data, with a focus on two key aspects: capturing the heterogeneity of tumor behavior and examining the spatial distribution of these behaviors within the tumor microenvironment. In the revised manuscript, we will clarify the scope of our study, emphasizing its limitations as an analytical tool rather than an image-processing solution. Additionally, we will provide references to relevant literature on available (open-source) software options for image processing (e.g. Diego Ulisse Pizzagalli et al J Immunol (2022); Aby Joseph et al eLife (2020) ;Molina-Moreno M et al Medical Image Analysis (2022); Hidalgo-Cenalmor, I et al, Nat Methods (2024); Ershov. D et al Nat Methods (2022)).

Regarding the reviewer's comment on our use of Imaris, we acknowledge that Imaris is a costly commercial software. However, based on our experience, it is widely used by the intravital microscopy community due to its user-friendly interface for 3D image visualization and analysis. Despite its limitations in accuracy and the fact that it is not open-source, we believe that including data processed with Imaris will be valuable to the IVM community.

However, to improve compatibility with data from other segmentation and tracking pipelines, we have already updated our tool to support formats generated by open-source Fiji plugins like TrackMate. These updates are available in our GitHub repository, and we will describe this functionality in detail in the revised manuscript to ensure compatibility with segmented and tracked data from various open-source platforms.

Comment: The number of cells (e.g. per behavioural cluster), and the number of independent mice, represented in each result figure, is not included in the figure legends and are difficult to ascertain from the methods.

We appreciate the reviewer's constructive feedback regarding the clarity of the number and type of replicates used in our analyses. In the revised manuscript, we will include detailed information in the figure legends regarding the number of cells (e.g., per behavioral cluster) and the number of independent mice represented in each result figure to ensure transparency.

Comment: The data used to test the pipeline in this manuscript is currently not available, making it difficult to assess its usability. It would be important to include this for researchers to use as a 'training dataset'.

As stated above we acknowledge that this was an oversight on our part and thank the reviewer for pointing this out. To address this, we have now added demo data to our GitHub repository (https://github.com/imAlgene-Dream3D/BEHAV3D_Tumor_Profiler/tree/BEHAV3D_TP-v2.0/demo_datasets). In the upcoming revised manuscript, we will also make sure to reference this addition. Additionally, we intend to provide both original and processed IVM movie samples to support users in navigating the complete pipeline effectively.

Comment: Precisely how the BEHAV3D-TP large-scale phenotyping module can map large-scale spatial phenotyping data generated using LSR-3D imaging data and Cytomap to 3D intravital imaging movies is unclear. Further details in the text and methods would be beneficial to aid understanding.

We appreciate the reviewer's comment and will provide additional details in the text and methods of the revised manuscript to clarify how the BEHAV3D-TP module maps LSR-3D and Cytomap data to 3D intravital imaging movies.

Comment: The analysis provides only preliminary evidence in support of the authors' conclusions on DMG cell migratory behaviours and their relationship with components of the tumour microenvironment. Conclusions should therefore be tempered in the absence of additional experiments and controls.

We appreciate the reviewer's comment and acknowledge that our conclusions should be tempered due to the preliminary nature of our evidence. To be able to directly analyze the impact of the brain tumor microenvironment on cancer cell behavior, we will include a new set of analyses in the revised manuscript. Specifically, we will utilize BEHAV3D-TP to analyze existing IVM data from adult gliomas with and without macrophage depletion (Alieva et al, Scientific Reports, 2017; <https://doi.org/10.1038/s41598-017-07660-4>) to evaluate the differences in heterogeneous cell populations under these conditions. Since this analysis pertains to a different tumor type, we will revise our conclusions accordingly and emphasize the necessity for additional experiments and controls to further validate our findings on DMG cell migratory behaviors and their relationship with the tumor microenvironment.

Reviewer #2:

Comment: The strength of democratizing this kind of analysis is undercut by the reliance upon Imaris for segmentation, so it would be nice if this was changed to an open-source option for track generation.

As noted in our previous response to Reviewer #1, we would like to point out that although Imaris is a commercial software, it is widely used in the intravital microscopy (IVM) community due to its user-friendly interface. One of its key advantages, which we also utilized, is semi-automated data tracking that allows for manual corrections in 3D—a process that can be more challenging in other open-source software with less effective data visualization.

However, we recognize that enhancing our pipeline's compatibility with open-source options is important. To this end, we have already updated our tool to support data formats generated by open-source Fiji plugins like TrackMate, improving compatibility with various segmentation and tracking pipelines (https://github.com/imAIgene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input). We will describe these updates in the revised manuscript to clarify our study's scope and the available image processing options.

Comment: The main issue is with the interpretation of the biological data in Figure 3 where ANOVA was used to analyse the proportional distribution of different clusters. Firstly the n is not listed so it is unclear if this represents an n of 3 where each mouse is an individual or whether each track is being treated as a test unit. If the latter this is seriously flawed as these tracks can't be treated as independent. Also, a more appropriate test would be something like a Chi-squared test or Fisher's exact test. Also, no error bars are included on the stacked bar graphs making interpretation impossible. Ultimately this is severely flawed and also appears to show very small differences which may be statistically different but may not represent biologically important findings. This would need further study.

We appreciate the reviewer's insightful comments regarding the interpretation of the biological data in Figure 3. To clarify, each mouse serves as an independent unit in this analysis. We believe that ANOVA is the appropriate test for comparing the proportions of different behavioral signatures across the tumor microenvironment (TME) regions identified by large-scale phenotyping. However, we acknowledge that using a stacked bar plot may have been misleading. While a Chi-squared test could show differences in the distribution of behavioral signatures, it would not indicate which specific signatures are responsible for those differences. Therefore, in the revised manuscript, we will retain the ANOVA analysis but will represent the proportions using a bar chart that clearly illustrates multiple conditions for each behavioral cluster. We also appreciate the reviewer's concern regarding the transparency of our data. In the revised manuscript, we will include the number of replicates for all figures to enhance clarity and understanding.

Comment: Figure 4 has similar statistical issues in that the n is not listed and, again, it is unclear whether they are treating each cell track as independent which, again, would be inappropriate. The best practice for this type of data would be the use of super plots as outlined in Lord et al. (2020) JCI - SuperPlots: Communicating reproducibility and variability in cell biology.

We appreciate the reviewer's comments and suggestions regarding Figure 4. In the revised manuscript, we will clarify the number of replicates used and our approach to treating cell tracks as independent units. We will implement super-plots where appropriate, to enhance the communication of reproducibility and variability in our data.

Comment: The main issue that this raises is that the large-scale phenotyping module and the heterogeneity module appear designed to produce these statistical analyses that are used in these figures and, if they are based on the assumption that each track is independent, then this will produce inappropriate analyses as a default.

We appreciate the reviewer's comment, though we find ourselves unsure about the specific concern being raised. To clarify, each mouse is treated as an independent unit in our analyses. For each large-scale phenotyping region, we measure the proportion of tumor cells displaying a specific behavioral phenotype independently for each mouse. These proportions are then used for statistical analysis. We hope this explanation provides clarity, and we will adjust the manuscript to better convey this methodology.

Reviewer #3:

Comment: The most challenging task of analyzing 3D time-lapse imaging data is to accurately segment and track the individual cells in 3D over a long time duration. BEHAV3D Tumor Profiler did not provide any new advancement in this regard, and instead relies on commercial software, Imaris, for this critical step. Imaris is known to have a very high error rate when used for analyzing 3D time-lapse data. In the Methods section, the authors themselves stated that "Tumor cell tracks were manually corrected to ensure accurate tracking". Based on our own experience of using Imaris, such manual correction is tedious and often required for every time step of the movie. Therefore, Imaris is not a satisfactory tool for analyzing 3D time-lapse data. Moreover, Imaris is expensive and many research labs probably can't afford to buy it. The fact that BEHAV3D Tumor Profiler critically depends on the faulty ImarisTrack module makes it unclear whether the BEHAV3D tool or the results are reliable.

If the authors want to "democratize the analysis of heterogeneous cancer cell behaviors", they should perform image segmentation and tracking using open-source codes (e.g., Cellpose, Stardisk & 3DCellTracker) and not rely on the expensive and inaccurate ImarisTrack Module for the image analysis step of BEHAV3D.

We appreciate the reviewer's comments on the challenges of segmenting and tracking individual cells in 3D time-lapse imaging data. As mentioned previously, our primary focus is to develop an analytical tool for comprehensive data analysis rather than developing tools for image processing. To enhance accessibility, we have updated our tool to support data formats from open-source Fiji plugins, such as TrackMate, which will benefit users without access to commercial software (https://github.com/imAIGene-Dream3D/BEHAV3D_Tumor_Profiler?tab=readme-ov-file#data-input).

While we recognize the limitations of Imaris, it remains widely used in the intravital microscopy community due to its user-friendly interface for 3D visualization and semi-automated segmentation capabilities. Since no perfect tracking method currently exist, we utilized Imaris for its ability to allow manual corrections of faulty tracks, ensuring the reliability of our results. This approach was the best available option when we began our analysis, allowing us to obtain accurate results efficiently.

In the revised manuscript, we will clarify our methodology and provide information on both Imaris and alternative processing options to strengthen the reliability of our findings.

Comment: The authors developed a "Heterogeneity module" to extract distinctive tumor migratory phenotypes from the cell tracks quantified by Imaris. The cell tracks of the individual tumor cells are all quite short, indicating relatively low motility of the tumor cells. It's unclear whether such short migratory tracks are sufficient to warrant the PCA

analysis to identify the 7 distinctive migratory phenotypes shown in Figure 2d. It's also unclear whether these 7 migratory phenotypes correspond to unique functional phenotypes.

For the 7 distinctive motility clusters, the authors should provide a more detailed analysis of the differences between them. It's unclear whether the difference in retreating, slow retreating, erratic, static, slow, slow invading, and invading correspond to functional difference of the tumor cells.

While some tumor cells exhibit limited motility, indicated by short tracks, others demonstrate significant migratory capabilities. This variability in tumor cell behavior is a central focus of our analysis, and our tool is specifically designed to identify and distinguish these differences. Our PCA analysis effectively captures this variability, as illustrated in Figure 2 d-f. It differentiates between cells exhibiting varying degrees of migratory behavior, including both highly migratory and less migratory phenotypes, as well as their directionality relative to the tumor core and the persistence of their movements. Thus, we believe that our approach provides valuable insights into the distinct migratory phenotypes within the tumor microenvironment. We will clarify these aspects further in the revised manuscript to enhance the reader's understanding of our findings.

While our current manuscript does not provide explicit evidence linking each motility cluster to functional differences among the tumor cells, it is important to note that the state of the field supports the idea that cell dynamics can predict cell states and phenotypes. Research conducted by ourselves (Dekkers, Alieva et al., Nat Biotech, 2023) and others, such as Craiciuc et al. (Nature, 2022) and Freckmann et al. (Nat Comm, 2022) has shown that variations in cell motility patterns are indicative of underlying functional characteristics. For instance, cell morphodynamic features have been shown to reflect differences in cell types, T cell targeting states, tumor metastatic potential, and drug resistance states. In the revised manuscript, we will reference relevant studies to underscore the biological significance of these behaviors. By doing so, we hope to clarify the potential implications of our findings and strengthen the overall narrative of our research.

Comment: Using only motility to classify tumor cell behaviours in the tumor microenvironment (TME) is probably not sufficient to capture the tumor cell difference. There are also other non-tumor cell types in the TME. If the authors aim to develop a computational tool that can elucidate tumor cell behaviors in the TME, they should consider other tumor cell features, e.g., morphology, proliferation state, and tumor cell interaction with other cell types, e.g., fibroblasts and distinct immune cells.

The authors should expand the scale of tumor behavior features to classify the tumor phenotype clusters, e.g., to include tumor morphology, proliferation state, and tumor cell interaction with other TME cell types.

We believe that using dynamic features alone is sufficient to capture differences in tumor behavior, as demonstrated by our results in Figure 2. However, we appreciate the reviewer's suggestion to consider additional features, such as cell morphology and interactions with other cell types, to finetune our analyses. To this end, we have adapted our pipeline to be compatible with various features present in the data (https://github.com/imAlgene-Dream3D/BEHAV3D_Tumor_Profiler/tree/BEHAV3D_TP-v2.0?tab=readme-ov-file#feature-selection). We will emphasize this in the revised manuscript. However, we would like to point out that not all features may provide informative insights and that a wide range of features can instead introduce biologically irrelevant noise, making interpretation more challenging. For instance, in 3D microscopy, the z-axis resolution is typically lower, which can lead to artifacts like elongation in that direction. Adding morphological features that capture this may skew the analysis. Therefore, we believe that incorporating additional features should be approached

with caution. We will clarify these considerations in the revised manuscript to better guide users in utilizing our computational tool effectively. We will also reference the use of unbiased feature selection techniques, such as bootstrapping methods, to identify biologically relevant features based on the conditions provided (D.G. Aragonés et al, Computers in Biology and Medicine (2024)).

Comment: The authors have already published two papers on BEHAV3D [Alieva M et al. Nat Protoc. 2024 Jul;19(7): 2052-2084; Dekkers JF, et al. Nat Biotechnol. 2023 Jan;41(1):60-69]. Although the previous two papers used BEHAV3D to analyze T cells, the basic pipeline and computational steps are similar, in particular regarding cell segmentation and tracking. The addition of a "Heterogeneity module" based on PCA analysis does not make a significant advancement in terms of image analysis and quantification.

We want to emphasize that we have no intention of duplicating our previous publications. In this manuscript, we have consistently cited our foundational papers, where BEHAV3D was first developed for T cell migratory analysis in *in vitro* settings. In the introduction, we clearly state that our earlier work inspired us to adopt a similar approach for analyzing cell behavior in intravital microscopy (IVM) data, addressing the specific needs and complexities of analyzing tumor cell behaviors in the tumor microenvironment.

Importantly, our new work provides several key advancements: 1) a pipeline specifically adapted for intravital microscopy (IVM) data; 2) integration of spatial characteristics from both large-scale and small-scale phenotyping; and 3) a zero-code approach designed to empower researchers without coding skills to effectively utilize the tool. We believe that these enhancements represent meaningful progress in the analysis of cell behaviors within the tumor microenvironment which will be valuable for the IVM community. We will ensure that these points are clearly articulated in the revised manuscript.

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